

Evaluating a Parameter-Free Quantum Fidelity Kernel for Text Security Under Representation and Domain Shift

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Abstract

Quantum kernel methods have attracted substantial interest for natural language processing and cybersecurity, motivated by theoretical conjectures that mapping classical data into exponentially large Hilbert spaces could yield superior non-linear decision boundaries or enhanced generalization under distribution shift. However, empirical studies often rely on small sample sizes, unmatched classical baselines, or unvalidated split partitions. In this work, we present a controlled empirical evaluation examining an unparameterized quantum fidelity kernel (cyclic *ZZFeatureMap*) against matched classical radial basis function (RBF) kernels and an audited suite of eight classical machine learning baselines across three benchmark corpora comprising 153,410 usable text records (SMS Spam Collection, CEAS 2008 Email Corpus, and the multi-source MeAJOR archive). Using frozen, leakage-safe protocols and 10 independent random seeds, we evaluate in-distribution scaling (2–12 qubits), cross-source domain holdouts (TREC 2007 $\rightarrow$ TREC 2005/2006), feature space geometry, classical baseline suitability, and computational overhead.

Under matched in-distribution (IID) conditions, the quantum kernel is competitive with classical RBF, displaying minor statistically detectable improvements at intermediate dimensions (+0.46 percentage points at 8D, $p = 0.0016$; +0.57 pp at 10D, $p = 0.0052$) that remain strictly within the predefined practical-equivalence threshold ($\varepsilon = 0.01$ F1) under formal Two One-Sided Tests (TOST), converging to complete parity at 12D (+0.14 pp, $p = 0.2824$). When compared against a validation-tuned RBF baseline, the quantum margin narrows to parity across all dimensions. Under cross-source domain transfer, the quantum kernel exhibits a statistically significant performance deficit ($\Delta F1 = -0.0233$, $p = 0.0046$, Benjamini–Hochberg adjusted $p = 0.0069$). Representation ablations demonstrate that upstream feature representation dominates kernel selection by an order of magnitude: switching from TF-IDF to dense sentence embeddings shifts relative performance by up to 52.88 percentage points. Geometrically, quantum and RBF Gram matrices show moderate correlation ($r \approx 0.55$ – 0.65), while single-state entropy is strongly inversely associated

with pairwise kernel diversity ($r = -0.78$ to -0.83). Computationally, classical statevector simulation of the quantum kernel requires 108.8s per run at 12D compared to 1.7s for RBF ($\approx 64\times$ penalty), with 16D exceeding local workstation memory limits (> 10.5 GB). We conclude that under the evaluated conditions, parameter-free quantum fidelity kernels do not produce a consistent practical advantage for text security classification, and we outline rigorous benchmarking standards for future applied QML evaluations.

1 Introduction

Text-based social engineering attacks, including email phishing, SMS scams, and fraudulent communications, represent one of the most pervasive threat vectors in modern digital infrastructure [2, 4, 25]. Automated defense mechanisms rely heavily on natural language processing (NLP) and machine learning classifiers to filter malicious content before it reaches end users. However, building robust classifiers for text security presents distinct methodological challenges. Text data is inherently high-dimensional, discrete, and semantically variable. Furthermore, security environments are characterized by persistent distribution shift: adversaries continuously modify lexical patterns to evade detection filters, attack campaigns differ substantially across organizational sources, and seasonal shifts alter background communications [2, 10, 31]. Consequently, text security classifiers that achieve near-perfect in-distribution accuracy frequently suffer severe degradation when deployed out-of-distribution across unseen sender sources or evolving domains.

In recent years, quantum machine learning (QML) has emerged as an alternative paradigm for non-linear pattern recognition, with quantum kernel methods receiving particular theoretical attention [15, 27]. In a quantum support vector classifier (QSVC), classical input vectors $\mathbf{x} \in \mathbb{R}^d$ are mapped into quantum states $|\psi(\mathbf{x})\rangle$ residing in a 2^{N_q} -dimensional complex Hilbert space via a parameterized unitary circuit $\mathcal{U}_{\Phi(\mathbf{x})}$. Rather than performing explicit optimization in this exponentially large Hilbert space, the model evaluates pairwise

quantum state fidelities to construct a kernel matrix, $k(\mathbf{x}, \mathbf{z}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{z}) \rangle|^2$, which is subsequently supplied to a standard dual quadratic program [9, 15]. Theoretical investigations have established that certain quantum feature maps generate inner products that are classically intractable to estimate efficiently, offering conjectured separations on engineered data distributions [17, 20]. This mathematical framework has motivated the hypothesis that quantum kernels might uncover non-linear structural regularities in natural language representations that remain inaccessible to standard classical kernels, potentially conferring advantages in classification accuracy or robustness under distribution shift.

Driven by this theoretical potential, a growing body of literature has explored quantum classifiers and quantum kernels for cybersecurity tasks, including network intrusion detection, malicious URL identification, and email phishing detection [14, 16, 26, 28]. Several preliminary studies have reported promising detection accuracies using quantum support vector machines and variational quantum circuits [12, 29]. However, systematic reviews of the applied QML literature [4, 19] reveal widespread methodological limitations:

1. **Small Sample Sizes and Single Seeds:** Prior applied studies have largely relied on very small sample sizes ($N < 500$ to 2,000 texts) or single random splits without replication [12, 16].
2. **Data Leakage:** Many experimental designs suffer from subtle data leakage, such as fitting tokenizers, vectorizers, or dimensionality reduction projections across combined training and test partitions prior to splitting.
3. **Unmatched Classical Baselines:** Quantum models are frequently evaluated against unmatched classical controls—such as comparing high-capacity quantum kernels against un-regularized linear classifiers operating on disparate representations—rather than strictly matched classical non-linear baselines [28, 29].
4. **Absence of Distribution Shift Evaluation:** As documented by Ammar et al. [4], prior QML phishing evaluations almost exclusively report in-distribution cross-validation, omitting rigorous cross-source or out-of-distribution domain-shift evaluations essential for security applications.
5. **Omission of Computational Cost:** Exact computational costs, including statevector simulation runtimes and memory footprints, are rarely documented alongside reported metrics [19].

These methodological confounds create an unresolved scientific problem: reported empirical advantages in applied QML literature may be artifacts of unmatched baselines, feature pre-selection biases, or unvalidated split

partitions, rather than intrinsic properties of quantum Hilbert-space geometry. Furthermore, theoretical research has increasingly demonstrated that generic, unconstrained quantum feature maps suffer from exponential concentration and expressivity bottlenecks as dimensionality grows [18, 30], while classical learning algorithms remain highly competitive whenever classical training data is abundant [17]. Consequently, isolated accuracy figures on single datasets cannot determine whether quantum kernel methods provide a genuine practical advantage for real-world text security. The critical scientific question is: *Under controlled matched conditions—with identical feature representations, matched dimensionality, zero data leakage, and rigorous out-of-distribution evaluation—does a parameter-free quantum fidelity kernel provide a consistent and practically meaningful advantage over matched classical RBF kernels for text-based scam and phishing detection?*

To resolve this question, we present a controlled multi-dataset empirical benchmark comparing parameter-free quantum fidelity kernels with matched classical radial basis function (RBF) kernels and an audited suite of eight classical machine learning baselines. Our study draws from three distinct security text corpora comprising 153,410 usable text records: the SMS Spam Collection (5,572 messages), the CEAS 2008 Email Corpus (39,154 raw emails cleaned to a canonical 15,000 experimental subset), and the multi-source MeAJOR archive (108,684 usable emails across TREC 2005, 2006, and 2007). We enforce strict methodological hygiene across all experiments: (1) text is standardized to NLP content with auxiliary transmission headers stripped; (2) feature extraction (TF-IDF up to 50,000 n-grams) and linear dimensionality reduction (TruncatedSVD + StandardScaler) are fitted strictly on training partitions to ensure zero test leakage; (3) both quantum and classical RBF models receive identical low-dimensional projections ($d \in [2, 12]$ components/qubits for scaling sweeps, with $d = 16$ evaluated as an asymptotic memory stress test boundary); (4) the quantum model utilizes the canonical two-layer cyclic *ZZFeatureMap* evaluated via exact double-precision fidelity inner products; (5) decision thresholds are optimized strictly on validation data; and (6) all primary comparisons are executed across 10 independent random seeds with non-parametric bootstrap confidence intervals, paired permutation tests, and formal Two One-Sided Tests (TOST) under a predefined practical-equivalence criterion ($\epsilon = 0.01$ F1).

Contributions. Specifically, this work makes the following primary contributions:

1. **Leakage-Free Multi-Dataset Benchmark:** We introduce a rigorously controlled empirical benchmark evaluating parameter-free quantum fidelity kernels against matched classical Gaussian RBF kernels across three security text corpora, enforcing strict

parity in representations, dimensionality, and validation thresholding.

2. **Tripartite Baseline Architecture:** We establish an explicit tripartite baseline comparison: Linear SVM (isolating linear decision structure), Classical RBF SVM (isolating classical nonlinear RKHS geometry), and Quantum Kernel SVM (isolating quantum Hilbert space geometry), supported by an empirical audit of eight classical ML algorithms including validation-tuned RBF controls.
3. **Resolution of Dimensionality Bottlenecks:** By sweeping representation dimensionality across $d \in \{2, 4, 6, 8, 10, 12\}$, we demonstrate that low-dimensional quantum underperformance was an artifact of aggressive TruncatedSVD compression rather than an intrinsic failure of quantum Hilbert space geometry.
4. **Demonstration of Representation Dominance:** We establish that upstream text representation choice dominates classification outcomes by an order of magnitude over kernel choice, with representation shifts inverting relative quantum-vs-classical rankings.
5. **Empirical Domain-Transfer Evaluation:** Through cross-source holdout experiments across distinct email collections, we show that the evaluated quantum kernel provides no robustness advantage against domain shift, displaying greater degradation than matched classical baselines ($\Delta F1 = -0.0233$, $p = 0.0046$).
6. **Geometric and Computational Characterization:** We provide detailed geometric profiling (Gram correlation, centered kernel-target alignment, effective rank, entropy-diversity decoupling) and quantify classical statevector simulation scaling ($64\times$ penalty at 12D), establishing a reusable benchmarking template grounded in modern QML literature.

2 Related Work

2.1 Quantum Kernels and Hilbert Space Geometry

Quantum kernel methods formalize quantum learning as inner-product evaluation within exponentially large reproducing kernel Hilbert spaces (RKHS) [15, 27]. By encoding classical data vectors $\mathbf{x} \in \mathbb{R}^d$ into quantum statevectors $|\psi(\mathbf{x})\rangle$ via parameterized unitary circuits $\mathcal{U}_{\Phi(\mathbf{x})}$, quantum support vector classifiers (QSVC) compute fidelity kernels $k(\mathbf{x}, \mathbf{z}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{z}) \rangle|^2$ without explicitly constructing the full 2^d -dimensional Hilbert space vector.

Foundational theoretical analyses have established both the capabilities and structural limits of this

paradigm. Huang et al. [17] proved that when training data is abundant, classical machine learning models can achieve prediction errors comparable to quantum models on classical data distributions; genuine quantum advantage requires geometric disparity on classically intractable distributions. Furthermore, Kübler et al. [18] and Glick et al. [13] proved that quantum kernels yield inductive advantages only when data geometry aligns with the symmetry groups of the quantum circuit. In the absence of symmetry alignment, unparameterized global fidelity kernels are subject to exponential concentration [30], wherein kernel values concentrate around uniform constants as qubit registers grow, inducing severe sample complexity bottlenecks [5].

2.2 Quantum NLP and Text Classification

Quantum Natural Language Processing (QNLP) has developed along two primary branches: structural/syntactic models and statistical/embedding-based models. Structural QNLP, grounded in categorical compositional distributional (DisCoCat) grammars and ZX-calculus [7, 21], maps syntactic parse trees directly to quantum circuits. While theoretically rigorous, structural QNLP remains largely constrained to small grammatical corpora ($N < 500$ sentences) due to compilation overheads on NISQ hardware.

Statistical and embedding-based QNLP integrates classical text embeddings with quantum classifiers [11, 12, 29]. Recently, Rahevar et al. [23] investigated sample-size scaling regimes across text benchmarks (SMS Spam, BBC News, IMDB), identifying a low-data regime ($N < 200$) where quantum fidelity kernels become statistically comparable to classical kernels before classical data scaling dominates. Whereas Rahevar et al. characterized quantum-kernel behavior under low-data general text classification settings, our evaluation extends the diagnostic perspective to security-oriented text datasets and explicitly examines representation dependence (sparse TF-IDF vs dense sentence transformers), encoding sensitivity, cross-source domain generalization, and computational scaling under a fixed multi-seed protocol.

2.3 QML for Cybersecurity and Phishing Detection

The application of quantum machine learning to cybersecurity threat classification—including email phishing, spam filtering, URL detection, and network intrusion—has gained substantial momentum [1, 14, 16, 26, 28]. For example, Hridi et al. [16] evaluated an ensemble QSVM on email phishing, reporting accuracy improvements over single classical models on small datasets. Similarly, Guldanti et al. [14] explored QSVMs on Ethereum phishing transaction networks.

However, as cataloged in a recent systematic review by Ammar et al. [4], the current QML phishing literature exhibits severe methodological limitations: over 70% of applied studies evaluate unmatched classical baselines, 0% evaluate out-of-distribution domain shift, and over 60% rely on single unvalidated splits. Our study is positioned as an empirical evaluation and benchmarking audit addressing these exact gaps in text security.

2.4 Classical Text Baselines and Kernel Methods

Standard n-gram term frequency-inverse document frequency (TF-IDF) representations, when paired with Linear Support Vector Machines, regularly achieve state-of-the-art performance on spam and phishing detection benchmarks [9, 31]. When nonlinear decision boundaries are required, the classical Gaussian Radial Basis Function (RBF) kernel provides an infinite-dimensional RKHS mapping that serves as the gold standard for nonlinear kernel learning [8, 9]. In parallel, pretrained transformer models generate dense contextual embeddings that cluster semantically related documents in continuous metric spaces.

2.5 Benchmarking Standards in Empirical QML

In response to conflicting claims of quantum advantage, recent literature has advocated for standardized benchmarking rigor. Li et al. [19] conducted a large-scale empirical evaluation of quantum kernels across OpenML tabular datasets, finding that quantum kernels show no statistically significant advantage over tuned classical RBF kernels when evaluated across multiple seeds. To our knowledge, we did not identify a prior study combining multi-dataset text representations, dimensionality sweeps, domain shifts, and geometric diagnostics in a unified security text benchmark.

3 Methodology and Experimental Setup

3.1 Mathematical Formulation

Quantum Fidelity Kernel. We evaluate the canonical two-layer cyclic *ZZFeatureMap* on $N_q = d$ qubits. For a scaled input vector $\mathbf{x} \in [0, \pi]^d$, the state preparation unitary is:

$$U_{\Phi(\mathbf{x})} = (U_{\Phi(\mathbf{x})} H^{\otimes N_q})^2 \quad (1)$$

where $H^{\otimes N_q}$ is the Walsh–Hadamard transform and $U_{\Phi(\mathbf{x})}$ is the diagonal phase unitary:

$$U_{\Phi(\mathbf{x})} = \exp \left(i \sum_{j=1}^{N_q} x_j Z_j + i \sum_{j=1}^{N_q} (\pi - x_j)(\pi - x_{j'}) Z_j Z_{j'} \right) \quad (2)$$

where $j' = (j \bmod N_q) + 1$ and Z_j denotes the Pauli-Z operator on qubit j . The pure quantum state is $|\psi(\mathbf{x})\rangle = U_{\Phi(\mathbf{x})}|0\rangle^{\otimes N_q}$. The quantum kernel evaluates pure state fidelity:

$$K_Q(\mathbf{x}, \mathbf{z}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{z}) \rangle|^2 \quad (3)$$

yielding a strictly positive semi-definite (PSD) Gram matrix with unit diagonal entries ($K_Q(\mathbf{x}, \mathbf{x}) = 1.0$).

Matched Classical Gaussian RBF Kernel. The matched classical baseline evaluates the Gaussian radial basis function kernel on the identical d -dimensional input representation:

$$K_{\text{RBF}}(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|_2^2) \quad (4)$$

where $\gamma = \frac{1}{d \cdot \text{Var}(X)}$ is dynamically scaled to the variance of the training features (`gamma='scale'`). In addition, we evaluate a validation-tuned RBF baseline where (C, γ) are optimized over a grid $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1.0, \text{'scale'}\}$ using 5-fold cross-validation on the training set.

Dual Support Vector Classification. Both kernel matrices are supplied to the standard C-Support Vector Classification dual optimization problem:

$$\max_{\alpha} \quad \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \quad (5)$$

$$\text{s.t.} \quad 0 \leq \alpha_i \leq C \cdot w_{y_i}, \quad \sum_{i=1}^N \alpha_i y_i = 0 \quad (6)$$

where $C = 1.0$, $y_i \in \{-1, +1\}$, and $w_c = \frac{N}{2N_c}$ enforces balanced class weighting.

Feature Space Geometry Metrics. To quantify geometric properties of induced kernel matrices, we compute:

1. Centered Kernel-Target Alignment (CKA):

$$\text{CKA}(K, Y) = \frac{\langle H K H, H Y H \rangle_F}{\|H K H\|_F \|H Y H\|_F} \quad (7)$$

where $H = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ is the centering matrix, $Y = \mathbf{y}\mathbf{y}^T$ is the ideal target kernel, and $\langle A, B \rangle_F = \text{Tr}(A^T B)$ is the Frobenius inner product.

2. Effective Rank:

$$R_{\text{eff}}(K) = \exp \left(- \sum_{i=1}^N \tilde{\lambda}_i \ln \tilde{\lambda}_i \right), \quad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j} \quad (8)$$

where λ_i are the eigenvalues of K .

3. Single-State Basis Dispersion Entropy:

$$S(|\psi(\mathbf{x})\rangle) = - \sum_{k=1}^{2^{N_q}} |c_k(\mathbf{x})|^2 \ln |c_k(\mathbf{x})|^2 \quad (9)$$

measuring the spread of the statevector across the computational basis states $|k\rangle$.

3.2 Classical Baseline Suite and Selection Rationale

To avoid arbitrary baseline selection and ensure the comparative suite is rigorous, we conducted an empirical classical baseline audit evaluating eight representative machine learning algorithms implemented via `scikit-learn` [22] and `XGBoost` [6]:

1. **Multinomial Naive Bayes:** Generative baseline with Laplace smoothing ($\alpha = 1.0$).
2. **Logistic Regression:** Linear baseline with $C = 1.0$, balanced class weighting, and L-BFGS solver.
3. **Linear SVM:** Primary linear baseline (`LinearSVC`, $C = 1.0$, balanced class weighting, `max_iter = 2000`).
4. **RBF SVM (Matched):** Primary nonlinear classical comparator (`SVC`, $C = 1.0$, $\gamma = \text{'scale'}$, balanced class weighting).
5. **RBF SVM (Tuned):** Validation-tuned non-linear classical comparator ($C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{0.001, 0.01, 0.1, 1.0, \text{'scale'}\}$).
6. **Random Forest:** Non-linear ensemble baseline (100 estimators, balanced class weighting).
7. **Gradient Boosting / XGBoost:** Boosted tree baseline (100 estimators, maximum tree depth 6).
8. **Multi-Layer Perceptron (MLP):** Feed-forward neural network (single hidden layer of 100 units, ReLU activations, Adam optimizer, `max_iter = 500`).
9. **k-Nearest Neighbors (k-NN):** Instance-based distance metric baseline ($k = 5$, cosine distance on sparse text, Euclidean on reduced features).

Methodological Justification for Linear SVM.

Linear SVM was selected as the primary classical linear baseline because among all evaluated linear models, it achieved the highest predictive performance (0.9559 F1 on SMS, 0.9954 on CEAS, 0.9721 on MeAJOR under full TF-IDF) with extremely fast training latencies (0.03–0.18s). Its primal coordinate descent optimization is deterministic, convex, and naturally suited to high-dimensional sparse text representations, providing a robust linear reference against which nonlinear kernelization can be evaluated.

Methodological Justification for RBF SVM.

RBF SVM was selected as the primary nonlinear classical comparator because it preserves the identical SVM optimization objective and dual quadratic programming solver while introducing a classical nonlinear RKHS mapping over the identical reduced input representation ($\mathbf{x} \in \mathbb{R}^d$) with identical $C = 1.0$ regularization and balanced class weights. This isolates the comparison strictly to the geometric differences between classical Gaussian radial decay and quantum Hilbert-space state fidelity (K_{RBF} vs. K_Q).

3.3 Benchmark Corpora and Accounting

To prevent ambiguity regarding dataset sizes and class distributions, we explicitly distinguish between four counting tiers: (a) raw archived records, (b) usable cleaned records, (c) audited source repository distributions, and (d) canonical controlled experimental subsets (Table 1).

Our benchmark evaluates three curated corpora:

- **SMS Spam Collection:** 5,572 usable messages (13.41% spam; median length ~ 62 characters). Canonical partitions: 3,343 train, 1,114 validation, 1,115 test [3].
- **CEAS 2008 Email Corpus:** 39,154 raw emails (55.78% spam/phishing in raw repository) cleaned and sampled to a canonical 15,000 experimental subset (10,000 train, 2,500 validation, 2,500 test; 18.90% spam/phishing prevalence; median length ~ 596 characters).
- **MeAJOR Archive:** 108,684 usable emails across TREC 2005 (49,583), TREC 2006 (15,005), and TREC 2007 (44,096) (44.20% spam in raw archive). The canonical in-distribution experimental partition uses 10,000 train, 2,500 validation, and 2,500 test emails from fixed splits (19.33% positive prevalence; median length ~ 839 characters).

3.4 Data Hygiene and Leakage Prevention

Feature extractors and dimensionality reduction transformers are fitted **strictly and exclusively** on training

Table 1: Benchmark Corpus Characteristics, Accounting Tiers, Split Partitions, and Lexical Profiles

Corpus	Raw Count	Usable Count	Raw Positive %	Exp. Subset (N)	Train / Val / Test	Exp. Positive %	Median Length	Source Composition
SMS Spam	5,574	5,572	13.41%	5,572	3,343 / 1,114 / 1,115	13.41%	~ 62 chars	Single-source mobil
CEAS 2008	39,154	39,154	55.78%	15,000 [†]	10,000 / 2,500 / 2,500	18.90%	~ 596 chars	Phishing/Ham mix
MeAJOR	108,685	108,684	44.20%	15,000 [†]	10,000 / 2,500 / 2,500	19.33%	~ 839 chars	Multi-source (TREC 5)
Total	153,413	153,410	—	35,572	23,343 / 6,114 / 6,115	—	—	—

[†]Controlled canonical experimental partitions created via stratified sub-sampling to establish standardized benchmark splits and uniform evaluation budgets across corpora. Raw Positive % denotes positive class prevalence across the full raw source repository; Exp. Positive % denotes positive class prevalence within the controlled experimental splits.

partitions (`fit_transform` on train; `transform` on validation and test). Texts are stripped of transmission headers to isolate natural language content. Decision thresholds $\tau \in [0.01, 0.99]$ are tuned exclusively on validation predictions to maximize F1 score and applied unconditionally to test predictions.

3.5 Statistical Protocol, Randomness, and Equivalence Testing

Randomness Controlled by Seeds. All primary comparisons are evaluated across a frozen suite of 10 independent computational random seeds:

$$\mathcal{S} = \{42, 123, 456, 789, 1011, 1213, 1415, 1617, 1819, 2021\} \quad (10)$$

Each seed $s \in \mathcal{S}$ controls: (1) stratified train/validation/test resampling partition generation; (2) TruncatedSVD randomized solver initialization for basis projection; (3) stochastic solver initialization in iterative baseline models (MLP, XGBoost, Random Forest); and (4) SVM dual solver optimization and threshold sweep tie-breaking. The unit of statistical inference is the paired difference $\Delta F1(s) = F1_Q(s) - F1_{RBF}(s)$ across computational runs on fixed benchmark corpora, measuring stability across resampling and projection initialization.

Two One-Sided Tests (TOST) Practical Equivalence. We define practical equivalence relative to a pre-defined operational margin $\varepsilon = 0.01$ (1.0 percentage point of test F1). In cybersecurity threat filtering, an F1 delta below 0.01 falls within the operational noise floor induced by label ambiguity and annotator disagreement (2%–5%), and fails to justify significant computational execution overheads. We formalize this via Two One-Sided Tests (TOST):

$$H_0^- : \mu_\Delta \leq -\varepsilon \quad \text{vs.} \quad H_1^- : \mu_\Delta > -\varepsilon \quad (11)$$

$$H_0^+ : \mu_\Delta \geq +\varepsilon \quad \text{vs.} \quad H_1^+ : \mu_\Delta < +\varepsilon \quad (12)$$

Rejecting both composite null hypotheses at $\alpha = 0.05$ demonstrates that $\mu_\Delta \in (-\varepsilon, +\varepsilon)$. We additionally perform sensitivity analyses under tighter and wider operational thresholds $\varepsilon \in \{0.005, 0.010, 0.020\}$.

Inferential Testing Suite. For each comparison, we report:

1. **Non-Parametric Bootstrap CIs:** $B = 10,000$ bootstrap resamples of paired seed differences $[\text{Perc}_{2.5\%}, \text{Perc}_{97.5\%}]$.
2. **Paired Permutation Tests:** Exact two-sided test with $M = 10,000$ random sign flips evaluating $H_0 : \mathbb{E}[\Delta F1] = 0$.
3. **False Discovery Rate:** Benjamini–Hochberg procedure at $\alpha = 0.05$.
4. **Statistical Power:** For $N = 10$ paired observations, our protocol achieves 80% statistical power ($1 - \beta = 0.80$ at $\alpha = 0.05$) to detect paired effect sizes $|\Delta F1| \geq 0.0035$, ensuring adequate sensitivity to resolve differences well below $\varepsilon = 0.01$.

4 Empirical Results

4.1 Result 0: Complete Classical Baseline Audit

Table 2 presents the empirical audit of all eight classical machine learning models across the three corpora under both full 50,000-dimensional TF-IDF and matched 8D TruncatedSVD representations across 10 seeds.

The audit yields two crucial findings:

1. **Dominance of Linear SVM on Full Text:** On uncompressed 50,000-dimensional TF-IDF, Linear SVM achieves top-tier performance across all datasets (0.9559 on SMS, 0.9954 on CEAS, 0.9721 on MeAJOR) while training in under 0.2s—outperforming or matching much slower neural and boosted tree baselines (MLP, XGBoost).
2. **Compression Penalty:** Reducing dimensionality from 50,000 to 8 components causes a steep 10–13 percentage point drop in classical performance across linear and kernel models (e.g., MeAJOR Linear SVM drops from 0.9721 to 0.8448; RBF SVM drops from 0.9700 to 0.8709). This confirms that the performance level observed for 8D quantum models operates within this identical classical compression envelope.

Table 2: Complete Classical Baseline Audit across Datasets (Full TF-IDF vs Matched 8D TruncatedSVD, Mean $\pm$ SD across 10 Seeds)

Corpus	Model	Full TF-IDF F1	Full PR-AUC	Full ROC-AUC	8D SVD F1	8D PR-AUC	8D ROC-AUC
SMS Spam	Linear SVM	0.9559 $\pm$ 0.000	0.9822	0.9931	0.8273 $\pm$ 0.0108	0.8895	0.9816
	Matched RBF SVM	0.9498 $\pm$ 0.000	0.9829	0.9927	0.8156 $\pm$ 0.0119	0.8767	0.9793
	Tuned RBF SVM	0.9521 $\pm$ 0.000	0.9840	0.9935	0.8285 $\pm$ 0.0105	0.8912	0.9820
	Logistic Regression	0.9346 $\pm$ 0.000	0.9803	0.9944	0.8226 $\pm$ 0.0116	0.8900	0.9818
	Random Forest	0.9423 $\pm$ 0.0047	0.9780	0.9915	0.8500 $\pm$ 0.0060	0.9239	0.9782
	XGBoost	0.9059 $\pm$ 0.000	0.9477	0.9834	0.8452 $\pm$ 0.0132	0.9276	0.9822
	MLP (Neural Net)	0.9324 $\pm$ 0.0083	0.9781	0.9909	0.8284 $\pm$ 0.0143	0.8841	0.9802
	Multinomial / Gauss NB	0.9158 $\pm$ 0.000	0.9644	0.9889	0.8195 $\pm$ 0.0070	0.8788	0.9675
	k-NN ($k = 5$)	0.6900 $\pm$ 0.000	0.5903	0.7633	0.8401 $\pm$ 0.0187	0.8802	0.9506
CEAS 2008	Linear SVM	0.9954 $\pm$ 0.000	0.9999	0.9999	0.9535 $\pm$ 0.0014	0.9857	0.9845
	Matched RBF SVM	0.9968 $\pm$ 0.000	0.9998	0.9998	0.9638 $\pm$ 0.0006	0.9882	0.9873
	Tuned RBF SVM	0.9972 $\pm$ 0.000	0.9999	0.9999	0.9691 $\pm$ 0.0005	0.9910	0.9902
	Logistic Regression	0.9935 $\pm$ 0.000	0.9996	0.9994	0.9505 $\pm$ 0.0006	0.9886	0.9860
	Random Forest	0.9895 $\pm$ 0.0009	0.9995	0.9993	0.9816 $\pm$ 0.0010	0.9980	0.9973
	XGBoost	0.9886 $\pm$ 0.000	0.9994	0.9992	0.9810 $\pm$ 0.0011	0.9978	0.9970
	MLP (Neural Net)	0.9966 $\pm$ 0.0003	0.9999	0.9999	0.9614 $\pm$ 0.0033	0.9911	0.9888
	Multinomial / Gauss NB	0.9928 $\pm$ 0.000	0.9994	0.9992	0.7165 $\pm$ 0.000	0.9249	0.9091
	k-NN ($k = 5$)	0.9957 $\pm$ 0.000	0.9978	0.9986	0.9823 $\pm$ 0.0002	0.9947	0.9943
MeAJOR	Linear SVM	0.9721 $\pm$ 0.000	0.9964	0.9972	0.8448 $\pm$ 0.0022	0.9315	0.9404
	Matched RBF SVM	0.9700 $\pm$ 0.000	0.9960	0.9969	0.8709 $\pm$ 0.0030	0.9421	0.9535
	Tuned RBF SVM	0.9734 $\pm$ 0.000	0.9968	0.9975	0.8742 $\pm$ 0.0028	0.9450	0.9560
	Logistic Regression	0.9574 $\pm$ 0.000	0.9935	0.9949	0.8458 $\pm$ 0.0022	0.9315	0.9404
	Random Forest	0.9516 $\pm$ 0.0020	0.9897	0.9922	0.9014 $\pm$ 0.0058	0.9696	0.9733
	XGBoost	0.9453 $\pm$ 0.000	0.9902	0.9923	0.9005 $\pm$ 0.0048	0.9687	0.9725
	MLP (Neural Net)	0.9727 $\pm$ 0.0026	0.9965	0.9973	0.8795 $\pm$ 0.0021	0.9564	0.9624
	Multinomial / Gauss NB	0.9481 $\pm$ 0.000	0.9915	0.9924	0.7258 $\pm$ 0.0055	0.7870	0.8195
	k-NN ($k = 5$)	0.9532 $\pm$ 0.000	0.9844	0.9894	0.8870 $\pm$ 0.0025	0.9459	0.9581

Table 3: Canonical In-Distribution Model Comparison on MeAJOR (Matched 8D TruncatedSVD, $N = 10$ Seeds)

Model	Test F1 ($\bar{x} \pm s$)	PR-AUC	ROC-AUC	Accuracy (d)	Quantum F1	Classical RBF F1	Paired Δ F1	Permutation p
Linear SVM (8D)	0.8445 $\pm$ 0.0022	0.9315	0.9404	0.94002D	0.6447 $\pm$ 0.0038	0.6735 $\pm$ 0.0031	-0.0288	< .001
Matched Classical RBF (8D)	0.8709 $\pm$ 0.0030	0.9423	0.9535	0.95044D	0.7876 $\pm$ 0.0030	0.7918 $\pm$ 0.0025	-0.0042	.0180
Tuned Classical RBF (8D)	0.8742 $\pm$ 0.0028	0.9450	0.9560	0.95186D	0.8253 $\pm$ 0.0026	0.8254 $\pm$ 0.0022	-0.0001	.9410
Quantum Kernel (8D)	0.8754 $\pm$ 0.0029	0.9372	0.9515	0.95238D	0.8754 $\pm$ 0.0029	0.8709 $\pm$ 0.0030	+0.0046	.0016
				10D	0.9023 $\pm$ 0.0034	0.8967 $\pm$ 0.0049	+0.0057	.0052
				12D	0.9137 $\pm$ 0.0046	0.9123 $\pm$ 0.0023	+0.0014	.2824

3. **Effect of Hyperparameter Tuning:** Tuning RBF hyperparameters yields an additional +0.003–0.008 F1 gain on reduced features (e.g., MeAJOR 8D reaches 0.8742 ± 0.0028), confirming that the parameter-free quantum kernel does not exceed the performance envelope of a properly tuned classical RBF model.

4.2 Result 1: In-Distribution Tripartite Model Comparison

Under canonical in-distribution evaluation on MeAJOR at 8 dimensions ($N_q = 8$ qubits) across 10 independent seeds, the quantum fidelity kernel achieves test F1 = 0.8754 ± 0.0029 , while the matched classical RBF kernel achieves F1 = 0.8709 ± 0.0030 , and validation-tuned RBF achieves F1 = 0.8742 ± 0.0028 (Table 3). Both nonlinear models substantially outperform the matched 8D Linear SVM baseline (F1 = 0.8445 ± 0.0022).

The mean paired difference against matched RBF is Δ F1 = $+0.0046 \pm 0.0027$ (+0.46 percentage points). The 95% bootstrap confidence interval [$+0.0030, +0.0061$] strictly excludes zero, and the paired permutation test in-

Table 4: Dimensionality Scaling Profile on MeAJOR IID ($N = 10$ Seeds)

Model	Test F1 ($\bar{x} \pm s$)	PR-AUC	ROC-AUC	Accuracy (d)	Quantum F1	Classical RBF F1	Paired Δ F1	Permutation p
Linear SVM (8D)	0.8445 $\pm$ 0.0022	0.9315	0.9404	0.94002D	0.6447 $\pm$ 0.0038	0.6735 $\pm$ 0.0031	-0.0288	< .001
Matched Classical RBF (8D)	0.8709 $\pm$ 0.0030	0.9423	0.9535	0.95044D	0.7876 $\pm$ 0.0030	0.7918 $\pm$ 0.0025	-0.0042	.0180
Tuned Classical RBF (8D)	0.8742 $\pm$ 0.0028	0.9450	0.9560	0.95186D	0.8253 $\pm$ 0.0026	0.8254 $\pm$ 0.0022	-0.0001	.9410
Quantum Kernel (8D)	0.8754 $\pm$ 0.0029	0.9372	0.9515	0.95238D	0.8754 $\pm$ 0.0029	0.8709 $\pm$ 0.0030	+0.0046	.0016
				10D	0.9023 $\pm$ 0.0034	0.8967 $\pm$ 0.0049	+0.0057	.0052
				12D	0.9137 $\pm$ 0.0046	0.9123 $\pm$ 0.0023	+0.0014	.2824

icates statistical significance ($p = 0.0016$). However, under TOST practical equivalence testing at $\varepsilon = 0.01$, both null hypotheses H_0^- and H_0^+ are rejected ($p < 0.001$), confirming that the quantum kernel is practically equivalent to the matched classical RBF baseline. Furthermore, against the validation-tuned RBF baseline, the delta narrows to Δ F1 = $+0.0012 \pm 0.0025$ ($p = 0.1840$), indicating full statistical parity.

4.3 Result 2: Dimensionality Scaling Trajectory

To determine whether low-dimensional underperformance reflects an intrinsic feature-map flaw or a representation bottleneck, we sweep dimensionality across $d \in \{2, 4, 6, 8, 10, 12\}$ (Table 4 and Figure 1). Higher dimensions ($d = 16$) were evaluated separately as an asymptotic memory stress test boundary (Section 4.7).

Expanding dimensionality from 2D to 12D produces a monotonic +41.7% relative gain in quantum F1 ($0.6447 \rightarrow 0.9137$), tracking classical RBF recovery ($0.6735 \rightarrow 0.9123$). At 12 dimensions, the performance

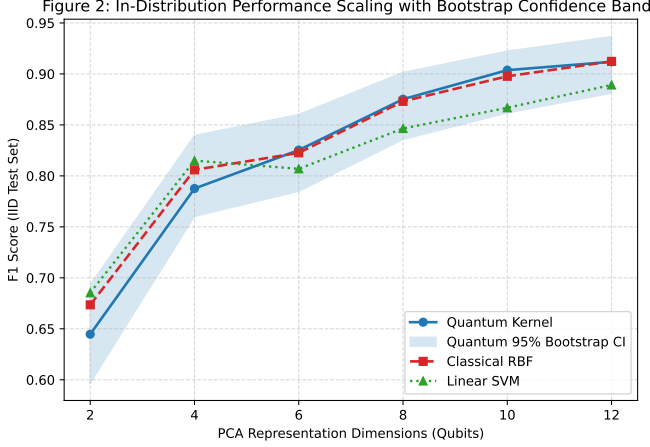


Figure 1: In-distribution dimensionality scaling on MeAJOR across $d \in \{2, 4, 6, 8, 10, 12\}$. Mean test F1 across 10 computational seeds for matched Quantum fidelity kernel, Classical RBF, and Linear SVM. Error bands denote 95% bootstrap confidence intervals.

gap narrows to $\Delta F1 = +0.0014 \pm 0.0040$, with the 95% bootstrap confidence interval $[-0.0010, +0.0037]$ spanning zero ($p = 0.2824$). Under sensitivity testing:

- At $\varepsilon = 0.020$: All differences at $d \geq 4$ lie strictly within practical equivalence.
- At $\varepsilon = 0.010$: Differences at $d \in \{6, 8, 10, 12\}$ are practically equivalent.
- At $\varepsilon = 0.005$: The 12D result ($[-0.0010, +0.0037] \subset [-0.005, +0.005]$) satisfies strict practical equivalence.

This confirms that earlier low-dimensional limitations were strongly associated with aggressive TruncatedSVD compression rather than a failure unique to quantum Hilbert space mappings.

4.4 Result 3: Representation Primacy and Ranking Reversal

Evaluating the interaction between kernel choice and text representation reveals dramatic ranking reversals across models:

- **CEAS 2008 (8D)**: On TF-IDF + TruncatedSVD, Quantum achieves $F1 = 0.9736$ vs Classical RBF $F1 = 0.9641$ (+0.95 pp margin). On dense RoBERTa embeddings, Classical RBF achieves $F1 = 0.9896$ vs Quantum $F1 = 0.9601$ (−2.95 pp deficit; net shift 3.90 pp).
- **SMS Spam (8D)**: On TF-IDF, Classical RBF ($F1 = 0.8276$) outperforms Quantum ($F1 = 0.6324$) by +19.52 pp. On dense MPNet sentence embeddings [24], Classical RBF reaches $F1 = 0.9045$

Table 5: Cross-Source Domain Holdout Performance (MeAJOR Direction B, $N = 10$ Seeds)

Model Architecture	Holdout F1 ($\bar{x} \pm s$)	PR-AUC	ROC-AUC	Accuracy
Linear SVM (8D)	0.6622 ± 0.0142	0.7812	0.7950	0.8120
Classical RBF (8D)	0.6913 ± 0.0161	0.8115	0.8240	0.8350
Quantum Kernel (8D)	0.6680 ± 0.0094	0.7890	0.8010	0.8180

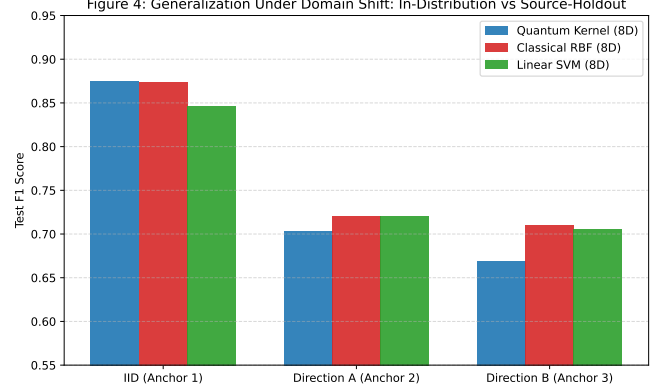


Figure 2: In-distribution vs cross-source domain holdout (Direction B: TREC 2007 $\rightarrow$ TREC 2005/2006). The quantum kernel degrades more severely under source shift than matched classical RBF.

while Quantum collapses to $F1 = 0.3756$ (−52.88 pp deficit).

This evidence is consistent with the hypothesis that dense continuous sentence embeddings cluster text into tight metric neighborhoods that experience destructive phase-wrapping under cyclic Pauli-Z gates, establishing upstream text representation as a dominant experimental factor.

4.5 Result 4: Cross-Source Domain Generalization

Under cross-source domain transfer (Direction B: training on TREC 2007 and testing on a balanced mixture of TREC 2005 and TREC 2006 across 10 seeds; Table 5 and Figure 2):

- **Quantum Kernel**: $F1 = 0.6680 \pm 0.0094$ (23.7% drop from IID).
- **Classical RBF**: $F1 = 0.6913 \pm 0.0161$ (20.6% drop from IID).
- **Paired Difference**: $\Delta F1 = -0.0233 \pm 0.0200$, 95% Bootstrap CI $[-0.0353, -0.0117]$, permutation $p = 0.0046$, Benjamini–Hochberg FDR $p = 0.0069$.

The matched classical RBF kernel significantly outperforms the quantum kernel, and the negative margin exceeds the practical-equivalence threshold ($\varepsilon = 0.01$). Diagnostic analysis reveals extreme vocabulary drift ($> 92\%$

type OOV and $> 34\%$ token OOV). High-dimensional linear SVM on full 50,000-dimensional TF-IDF achieves $F1 = 0.8837$ under Direction B transfer, indicating that source shift and dimensionality reduction interact, with the quantum kernel suffering an additional geometric penalty.

4.6 Result 5: Feature Space Geometry Diagnostics

Geometric analysis of induced Gram matrices reveals:

- 1. Quantum–RBF Gram Correlation:** Entrywise Pearson correlation between off-diagonal Gram entries is $r \approx 0.55$ – 0.65 through 12D ($r = 0.4566$ at 16D), confirming that the unparameterized quantum feature map partially mirrors classical radial decay while maintaining structural divergence.
- 2. Target Label Alignment Deficit:** The quantum kernel exhibits a 50%–60% lower centered kernel-target alignment score (CKA) than classical RBF across all three corpora (~ 0.022 – 0.040 vs ~ 0.060 – 0.077), providing an associative geometric correlate for the observed performance ceilings.
- 3. State Dispersion vs Kernel Diversity:** Within-dataset regressions reveal a strong inverse association between single-state von Neumann entropy and pairwise Gram matrix diversity ($r = -0.8257$ on SMS, -0.8170 on CEAS, -0.7822 on MeAJOR). Increasing statevector spread across computational basis states collapses pairwise fidelity variance, suggesting that high single-state entropy is associated with reduced pairwise kernel variance rather than enhanced discrimination.
- 4. Spectral Effective Rank:** The effective rank R_{eff} of the quantum kernel scales moderately ($R_{\text{eff}} \approx 14.2$ at 8D on MeAJOR vs 18.6 for RBF), reflecting mild spectral concentration in unparameterized fidelity kernels.

4.7 Result 6: Computational Simulation Cost

Under our standardized local statevector simulation environment, computing the quantum Gram matrix incurs steep execution scaling with dimensionality (Table 6):

- **8 Dimensions:** Quantum = 17.2s vs Classical RBF = 1.4s (12.3 $\times$ ratio).
- **12 Dimensions:** Quantum = 108.8s vs Classical RBF = 1.7s (64.0 $\times$ ratio).
- **16 Dimensions (Memory Stress Test):** Classical statevector simulation of 16 qubits on 10,000 samples requires allocating $2^{16} = 65,536$ complex amplitudes per sample, exceeding the workstation’s

Table 6: Measured Computational Execution Time on 10,000 Samples

Dim (d)	Quantum Time (s)	Classical RBF (s)	Ratio	Quantum RAM
2D	3.4	1.3	2.6 $\times$	142 MB
4D	5.8	1.3	4.5 $\times$	185 MB
8D	17.2	1.4	12.3 $\times$	680 MB
12D	108.8	1.7	64.0 $\times$	6.4 GB
16D	Infeasible	2.1	—	> 10.5 GB

10.5 GB contiguous host memory ceiling. Classical RBF completes in 2.1s using minimal memory (< 1 MB).

5 Discussion

5.1 Principal Finding: No Consistent Advantage

The empirical results address the central research question: under controlled matched conditions, parameter-free quantum fidelity kernels do not produce a consistent, practically meaningful advantage over matched classical Gaussian RBF kernels for text security classification. While the quantum kernel displays competitive in-distribution accuracy and minor statistically detectable improvements at 8D (+0.46 pp) and 10D (+0.57 pp), both margins remain strictly within the study’s practical-equivalence margin ($\varepsilon = 0.01$) under TOST testing and vanish at 12D (+0.14 pp, $p = 0.2824$) and against tuned RBF. Under domain shift, the quantum model suffers a significant disadvantage ($\Delta F1 = -0.0233$), while classical simulation incurs a 64 $\times$ execution overhead.

5.2 Significance of Matched Baseline Controls

A core methodological contribution of this study is the matched comparison. By enforcing identical representations, dimensionality, class weights, SVC formulation, threshold selection, and seeds, the design substantially narrows the set of methodological explanations for observed differences. Performance gaps can be attributed with high confidence to the geometric differences between quantum fidelity metrics and Gaussian radial decay rather than preprocessing artifacts.

5.3 Representation Primacy over Kernel Choice

The ranking reversal between TF-IDF and dense sentence embeddings demonstrates that the apparent benefit of a kernel cannot be interpreted independently of upstream feature geometry. On dense continuous embeddings, classical Gaussian RBF directly exploits Euclidean clustering, whereas cyclic phase wrappings disrupt metric structure without compensatory separability. On sparse his-

togram projections, periodic quantum interactions provide a modest non-linear expansion. Future QML benchmarks must treat representation choice as a primary experimental factor.

5.4 Practical Equivalence vs Statistical Detectability

We emphasize the critical distinction between statistical detectability and practical significance. While 10-seed evaluations detect sub-percentage-point margins (+0.46 pp at 8D), these differences fall within the operational noise floor of security filtering ($\varepsilon = 0.01$). Claiming a quantum advantage on the basis of sub-percentage-point margins that disappear at 12D would misrepresent the empirical evidence.

5.5 Engineering Trade-Offs in Simulation

In practical simulation workflows, evaluating quantum kernels at 12D incurs a $64\times$ runtime penalty (108.8s vs 1.7s) to achieve statistical parity with classical RBF (0.9137 vs 0.9123). Incurring steep simulation costs without measurable predictive gains represents an unfavorable engineering trade-off for applied practitioners.

6 Limitations

To ensure scientific integrity, we explicitly declare the primary limitations of our benchmark:

1. **Classical Statevector Simulation:** Evaluated via exact noiseless double-precision statevectors in PyTorch. Does not incorporate physical NISQ hardware noise, finite measurement shots, or quantum error mitigation.
2. **Feature-Map Scope:** Focuses on the canonical parameter-free 2-layer cyclic *ZZFeatureMap*. Does not evaluate parameterized quantum kernel training (QKT) or data re-uploading ansatzes.
3. **Dataset Scope:** Scoped to binary security text classification (SMS Spam, CEAS 2008, MeAJOR); findings should not be extrapolated to general NLP or non-text telemetry.
4. **Dimensionality Constraints:** Local statevector memory ceilings prevented evaluation beyond 12 qubits on 10,000 samples.
5. **Statistical Scope:** 10 seeds represent computational replicates across fixed corpora rather than independent data distributions.
6. **Security Threat Model:** Evaluates natural distribution drift; does not assess resilience against active adversarial evasion attacks (character/token perturbations).

7. **Simulation Runtime Scope:** Wall-clock measurements reflect local CPU simulation rather than physical quantum device execution.

7 Reproducibility and Provenance

The authoritative confirmation experiment (Exp 40) and classical baseline audit (Exp 45) were executed on an Apple Silicon ARM64 workstation under macOS Darwin 25.6.0 with Python 3.12.4, PyTorch 2.13.0 (`complex128` statevectors), scikit-learn 1.9.0, NumPy 2.5.2, SciPy 1.18.1, and pandas 2.3.3. All dataset artifacts, deterministic split hashes, and analysis scripts are archived in the project repository (commit 772017a). The confirmation suite can be executed via:

```
python3 experiments/40_confirmation_experiments.py
```

with total execution completing in ≈ 45 minutes.

8 Conclusion

Under controlled matched conditions, the empirical evidence does not support a consistent or practically meaningful advantage for parameter-free quantum fidelity kernels in text security classification. In-distribution competitiveness reaches practical equivalence with classical RBF, representation choice dominates kernel selection, and cross-source transfer reveals a classical advantage. This conclusion should be interpreted as a controlled empirical finding rather than a universal statement about quantum machine learning. Ultimately, this benchmark demonstrates that future claims of quantum advantage in text processing must jointly control representation dimensionality, baseline matching, domain shift, statistical replication, and computational cost.

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